(Follow Landau - Lifshitz, vol. 5, Ch. 59 "Magnetism of electron gas")

Pauli susceptibility of graphene.

P2 Very generally, start with

expression without magnetic field

$$\Omega = -k_B T \left\{ \int \frac{d^2\vec{p}}{(2\pi\hbar)^2} \ln\left(1 + e^{\beta(\mu - E_-(p))}\right) + \ln\left(1 + e^{\beta(\mu - E_+(p))}\right) \right\}$$

this accounts for $E_\pm(p)$ bands; $\beta \equiv 1/k_B T$.

$E_\pm(p) = \pm v_F |\vec{p}|$ near K and K' points.

$$\Omega_0(\mu) = -g_v g_s k_B T \int \frac{d^2\vec{p}}{(2\pi\hbar)^2} \left\{ \ln\left(1 + e^{\beta\mu} e^{\beta v_F |p|}\right) + \ln\left(1 + e^{\beta\mu} e^{-\beta v_F |p|}\right) \right\}$$

valley degeneracy ($g_v = 2$)

spin degeneracy ($g_s = 2$)

In magnetic field electron's energy changes due to the Zeeman interaction $\pm \mu_B H$, $\mu_B = \frac{|e|\hbar}{2mc}$, and can be accounted for by shifting the chemical potential $\mu \to \mu \pm \mu_B H$ for electrons with different spins.

Then $\Omega(\mu) = \frac{1}{2}\Omega_0(\mu + \mu_B H) + \frac{1}{2}\Omega_0(\mu - \mu_B H)$

where Ω_0 is thermodynamic potential in the absence of the field.

For weak field, when $\mu_B H \ll k_B T$, we can Taylor expand: $\Omega_0(\mu \pm \mu_B H) = \Omega_0(\mu) \pm \frac{\partial \Omega_0}{\partial \mu}(\mu_B H) + \frac{1}{2}(\mu_B H)^2 \frac{\partial^2 \Omega_0}{\partial \mu^2}$

use in there

$$\Omega(\mu) = \frac{1}{2}\left(\Omega_0 + \mu_B H \frac{\partial \Omega_0}{\partial \mu} + \frac{1}{2}(\mu_B H)^2 \frac{\partial^2 \Omega_0}{\partial \mu^2}\right) +$$

$$+ \frac{1}{2}\left(\Omega_0 - \mu_B H \frac{\partial \Omega_0}{\partial \mu} + \frac{1}{2}(\mu_B H)^2 \frac{\partial^2 \Omega_0}{\partial \mu^2}\right) =$$

$$= \Omega_0(\mu) + \frac{1}{2}(\mu_B H)^2 \frac{\partial^2 \Omega_0}{\partial \mu^2}$$

$$\frac{\partial \Omega_0}{\partial \mu} = -K_B T \underbrace{\int \frac{d\vec{p}}{(2\pi\hbar)^2}}_{g_v g_s}\left(\frac{\beta e^{\beta\mu} e^{\beta v_F p}}{1+e^{\beta\mu} e^{\beta v_F p}} + \frac{\beta e^{\beta\mu} e^{-\beta v_F p}}{1+e^{\beta\mu} e^{-\beta v_F \mu}}\right) =$$

$$= -g_v g_s \int \frac{d\vec{p}}{(2\pi\hbar)^2}\left(\frac{1}{e^{-\beta(v_F p+\mu)}+1} + \frac{1}{e^{\beta(v_F p-\mu)}+1}\right) =$$

$$= -g_v g_s \int \frac{d\vec{p}}{(2\pi\hbar)^2} \frac{e^{\beta(v_F p-\mu)}+1 + e^{-\beta(v_F p+\mu)}+1}{e^{-2\beta\mu} + e^{-\beta\mu}\left(e^{-\beta v_F p} + e^{\beta v_F p}\right)+1} =$$

$$= -g_v g_s \int \frac{d\vec{p}}{(2\pi\hbar)^2} \frac{e^{-\beta\mu}\cdot 2\,ch(\beta v_F p) + 2}{e^{-\beta\mu}\cdot 2\,ch(\beta\mu) + e^{-\beta\mu}\cdot 2\,ch(\beta v_F p)} =$$

$$= -g_v g_s \int \frac{d\vec{p}}{(2\pi\hbar)^2} \frac{ch(\beta v_F p) + e^{\beta\mu}}{ch(\beta v_F p) + ch(\beta\mu)} = -g_v g_s \int \frac{d\vec{p}}{(2\pi\hbar)^2} \frac{ch(\beta v_F p) + ch\beta\mu + sh\beta\mu}{ch(\beta v_F p) + ch(\beta\mu)}$$

$$\cancel{\frac{\partial^2 \Omega_0}{\partial \mu^2} = -g_v g_s \int \frac{d\vec{p}}{(2\pi\hbar)^2} \frac{\beta e^{\beta\mu}}{ch(\beta v_F p) + ch(\beta\mu)} - \frac{(ch(\beta v_F p) + e^{\beta\mu})\cdot \beta\, sh(\beta\mu)}{(ch(\beta v_F p) + ch(\beta\mu))^2}}$$

$$\cancel{= -g_v g_s \beta \int \frac{d\vec{p}}{(2\pi\hbar)^2} \frac{e^{\beta\mu}(ch(\beta v_F p) + ch(\beta\mu)) - (ch(\beta v_F p) + e^{\beta\mu})\, sh(\beta\mu)}{[ch(\beta v_F p) + ch(\beta\mu)]^2}}$$

$$\cancel{= -g_v g_s \int \frac{d\vec{p}}{(2\pi\hbar)^2} \frac{ch(\beta v_F p)(e^{\beta\mu} - sh\,\beta\mu) + e^{\beta\mu}(ch\,\beta\mu - sh\beta\mu)}{[ch(\beta v_F p) + ch(\beta\mu)]^2}}$$

Thus $$\frac{\partial \Omega_0}{\partial \mu} = -g_v g_s \int \frac{d\vec{p}}{(2\pi\hbar)^2}\left(1 + \frac{sh(\beta\mu)}{ch(\beta\mu) + ch(\beta v_F p)}\right)$$

$$\frac{\partial^2 \Omega_0}{\partial \mu^2} = -g_v g_s \int \frac{d\vec{p}}{(2\pi\hbar)^2}\left(\frac{\beta\, ch(\beta\mu)}{ch(\beta\mu) + ch(\beta v_F p)} - \frac{sh(\beta\mu)\cdot \beta\, sh(\beta\mu)}{\left(ch\beta\mu + ch(\beta v_F p)\right)^2}\right)$$

$$= -g_v g_s \beta \int \frac{d\vec{p}}{(2\pi\hbar)^2} \frac{ch^2(\beta\mu) - sh^2(\beta\mu) + ch(\beta\mu)\, ch(\beta v_F p)}{\left(ch(\beta\mu) + ch(\beta v_F p)\right)^2}$$

$$= \left(p = \frac{x}{\beta v_F}\right) \quad = -g_v g_s \cdot \frac{2\pi\beta}{(2\pi\hbar)^2} \frac{1}{(\beta v_F)^2} \int_0^\infty dx \cdot x \frac{ch^2(\beta\mu) - sh^2(\beta\mu) + ch(\beta\mu)\, ch\, x}{\left(ch(\beta\mu) + ch(x)\right)^2}$$

① Suppose $\mu = 0 \Rightarrow$ $$\frac{\partial^2 \Omega_0}{\partial \mu^2} = \frac{-g_v g_s k_B T}{2\pi(\hbar v_F)^2} \underbrace{\int_0^\infty dx \frac{x}{1 + ch\, x}}_{\ln(4)}$$

so that $$\Omega(\mu) = \Omega_0(\mu) - \frac{1}{2}\mu_B^2 H^2 \frac{g_v g_s \overset{\ln 4}{\checkmark} k_B T}{2\pi(\hbar v_F)^2}$$

$$M = -\frac{\partial \Omega}{\partial H} = \mu_B^2 H \frac{g_v g_s \ln(4)\, k_B T}{2\pi(\hbar v_F)^2}$$

Pauli suscept. $$\chi = \frac{\partial M}{\partial H} = \mu_B^2 \frac{g_v g_s \ln(4)}{2\pi(\hbar v_F)^2} k_B T$$

While Orbital susceptibility $$\chi_{orb} = -\frac{g_v g_s e^2 v_F^2}{6\pi c^2} \frac{1}{2 k_B T}$$

Compare magnitudes $$\left|\frac{\chi_{Pauli}}{\chi_{orb}}\right| = \frac{\mu_B^2 \ln(4) \cdot 6 (k_B T)^2 c^2}{e^2 v_F^2 \cdot \hbar^2 v_F^2}$$

$$\mu_B^2 = \left(\frac{e\hbar}{2mc}\right)^2$$

$$\left|\frac{\chi_P}{\chi_{orb}}\right| = \frac{e^2\hbar^2}{4m^2c^2}\,\frac{c^2}{e^2\hbar^2 v_F^4}\,(k_B T)^2\, 6\ln(4) =$$

$$= 3\ln(2)\left(\frac{k_B T}{m v_F^2}\right)^2$$

Estimate: at room $T \simeq 300\,K$, $k_B T = 1.4\cdot 10^{-23}\cdot 300\,J$
$\approx 5\cdot 10^{-21}\,J$

$$m v_F^2 \approx 10^{-30}\,kg\cdot(10^6\,m/s)^2 = 10^{18}\,J$$

$$\Rightarrow \left(\frac{k_B T}{m v_F^2}\right)^2 \sim (5\cdot 10^{-3})^2 \sim 10^{-5} \ll 1.$$

At $\mu=0$, Pauli susceptibility is negligible by comparison with the (Landau) orbital one.

Now $\mu \neq 0$ but $\mu\beta \gg 1$

$$\frac{\partial \Omega_0}{\partial \mu} \sim \frac{-\beta}{(v_F \hbar)^2}\left[\int_0^{\infty} dp\, p \; \frac{1 + ch(\beta\mu)\, ch(\beta p)}{(Ch\,\beta\mu + ch\,\beta p)^2}\right]$$

$$ch^2\mu - sh^2\mu = \frac{1}{4}\left(e^{\mu}+e^{-\mu}\right)^2 - \frac{1}{4}\left(e^{\mu}-e^{-\mu}\right)^2 = \frac{1}{4}\,4 = 1$$

$$= \int_0^{\mu} dp\, p \; \frac{1 + \frac{1}{4} e^{\beta\mu} e^{\beta p}}{\frac{1}{4}\left(e^{\beta\mu}+e^{\beta p}\right)^2} \sim e^{-\beta\mu}\int_0^{\mu} dp\, p\, e^{\beta p}$$

$$+ \int_{\mu}^{\infty} dp\, p \; \frac{1 + \frac{1}{4} e^{\beta\mu} e^{\beta p}}{\frac{1}{4} e^{2\beta p}} \sim e^{\beta\mu}\int_{\mu}^{\infty} dp\, p\, e^{-\beta p}$$

$$= \frac{1}{\beta^2}\left[e^{-\beta\mu}\, e^{\beta p}(\beta p - 1)\Big|_0^{\mu} + e^{\beta\mu}\left(-e^{-\beta p}\right)(\beta p + 1)\Big|_{\mu}^{\infty}\right]$$

$$= \frac{1}{\beta^2}\left[\beta\mu - 1 - e^{-\beta\mu}(-1) + e^{\beta\mu} e^{-\beta\mu}(\beta\mu + 1)\right]$$

$$= \frac{2\beta\mu}{\beta^2}$$

$$\Rightarrow \frac{\partial \Omega_0}{\partial \mu} \sim -\frac{\beta}{(\hbar v_F)^2}\,\frac{2\beta\mu}{\beta^2} \sim -\frac{\mu}{(\hbar v_F)^2}$$

$$\Rightarrow M \sim \mu_B^2 H \frac{\mu}{(\hbar v_F)^2} \text{ and } \chi_{Pauli} \sim \mu_B^2 \frac{\mu}{(\hbar v_F)^2}$$

while orbital susceptibility on this regime is exponentially small, $\chi_{orb} \sim \frac{1}{T} e^{-\beta\mu}$.